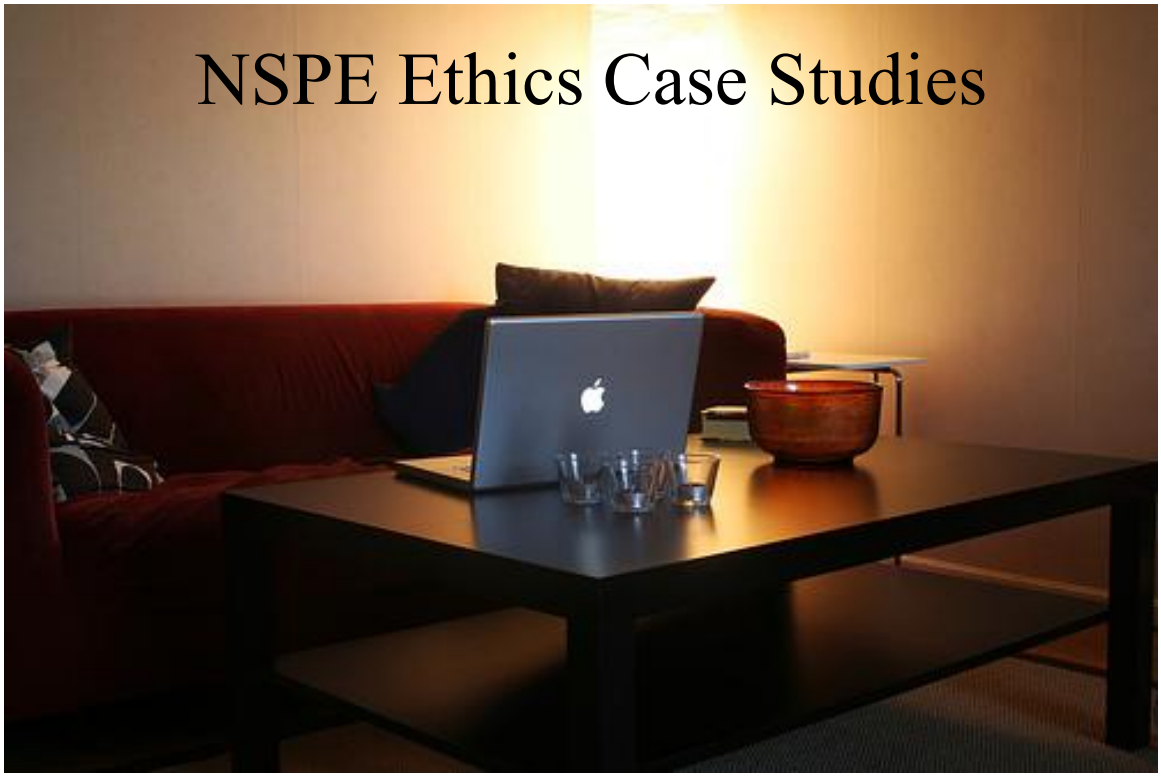


NSPE Ethics Case Studies

By

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1 PDH HOURS

NSPE ETHICS CASE STUDIES

Introduction

Engineers are expected to follow behavior standards (codes of conduct) set and enforced by:

- Federal, state, or local regulations (laws and rules)
- Employers (corporate policy, office standards, employment agreements, etc.)
- Professional organizations (NSPE, NCEES, ASCE, etc.)

The most widely adopted standard from a professional organization is called the "Code of Ethics for Engineers" (Code of Ethics) by the National Society of Professional Engineers (NSPE). This code includes six fundamental canons (principles of conduct), five rules of practice, and nine professional obligations, as outlined later in this course.

The NSPE Board of Ethical Review (BER) publishes ethical review cases, also called case studies, to exemplify how to apply the Code of Ethics to real world situations. The case studies are available on the NSPE website, as shown in Figure 1. The website is listed in the References section.

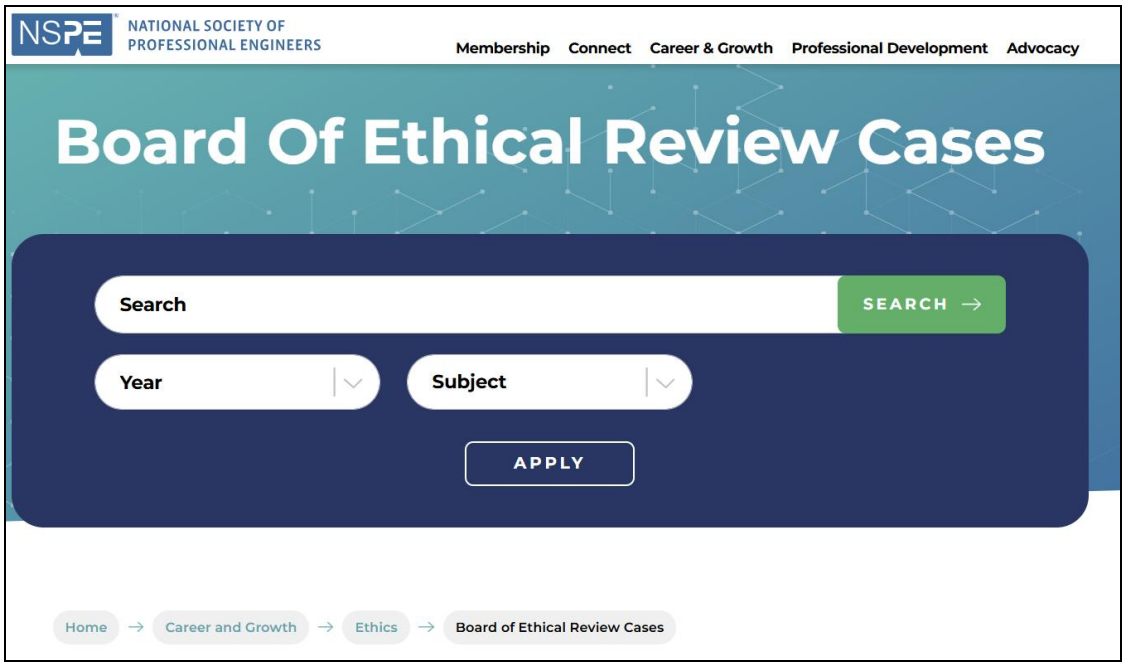


Figure 1: NSPE website with searchable database of BER ethics cases.

Source: www.nspe.org/career-growth/ethics/board-ethical-review-cases

NSPE Code of Ethics for Engineers

The following are the main statements of the NSPE Code of Ethics for Engineers, current as of October 2025. The full document with all subsections can be found on the website listed in the references section at the end of this course.



I. Fundamental Canons

Engineers, in the fulfillment of their professional duties, shall:

1. Hold paramount the safety, health, and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public.
2. Engineers shall perform services only in the areas of their competence.
3. Engineers shall issue public statements only in an objective and truthful manner.
4. Engineers shall act for each employer or client as faithful agents or trustees.
5. Engineers shall avoid deceptive acts.

III. Professional Obligations

1. Engineers shall be guided in all their relations by the highest standards of honesty and integrity.
2. Engineers shall at all times strive to serve the public interest.
3. Engineers shall avoid all conduct or practice that deceives the public.

4. Engineers shall not disclose, without consent, confidential information concerning the business affairs or technical processes of any present or former client or employer, or public body on which they serve.
5. Engineers shall not be influenced in their professional duties by conflicting interests.
6. Engineers shall not attempt to obtain employment or advancement or professional engagements by untruthfully criticizing other engineers, or by other improper or questionable methods.
7. Engineers shall not attempt to injure, maliciously or falsely, directly or indirectly, the professional reputation, prospects, practice, or employment of other engineers. Engineers who believe others are guilty of unethical or illegal practice shall present such information to the proper authority for action.
8. Engineers shall accept personal responsibility for their professional activities, provided, however, that engineers may seek indemnification for services arising out of their practice for other than gross negligence, where the engineer's interests cannot otherwise be protected.
9. Engineers shall give credit for engineering work to those to whom credit is due, and will recognize the proprietary interests of others.

Case 24-1: Balancing Client Directives and Public Welfare: Stormwater Management Dilemma

Date Published: July 18, 2025

Facts:

Engineer L, a licensed professional engineer, has many years of experience in stormwater control design. Engineer L is contracted by Client X, a private development entity, to design a stormwater management system for a large residential and commercial development near a small community. This community relies on a nearby surface water watershed as a primary drinking water source. Part of Engineer L's scope of work for Client X includes ensuring that stormwater from the new development will not impact this primary drinking water source for the community. During the preliminary design phase, Engineer L becomes concerned that the risk of impact to the community drinking water source will potentially increase, as the likelihood of heavy stormwater flows will increase over time. But before Engineer L can quantify the increased risk, Client X encounters unexpected financial setbacks and requests that Engineer L suspend work identified in the original contract. In their communications about the suspension, Engineer L does not mention to Client X the potential increased risk of impact to the community drinking water source.

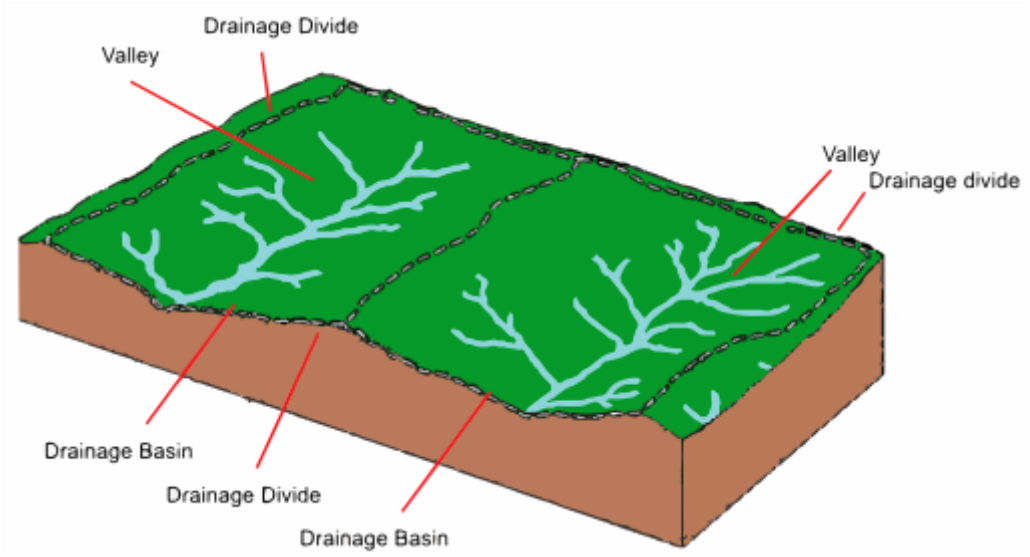


Figure 2: Example of two watersheds (drainage basins). In the case study, a community relays on making drinking water from one watershed, while a nearby site development creates a risk of contaminated runoff entering the watershed.

Source: commons.wikimedia.org/wiki/File:Figure_of_a_drainage_basin.svg

Several months later, Client X's financial situation improves and Engineer L is asked to resume work on the stormwater management system. About this same time, an historic

heavy rainfall in the region leads to increased stormwater accumulation on-site, raising the risk of runoff into the watershed. Engineer L conducts additional studies and qualitatively estimates the risk that heavy rainfall could lead to stormwater runoff from the development reaching the nearby watershed and community drinking water source. Engineer L notifies Client X of this risk, advising that additional protective measures should be considered, especially as local environmental standards require steps to safeguard public water sources. However, Client X is hesitant to invest in additional protective measures, citing continuing budget limitations. Despite Engineer L's concerns, Client X insists on proceeding without added safeguards, stating that Client X will address any compliance issues later, if needed.

Question 1: Was it ethical for Engineer L to cease work when requested by Client X, without voicing concern about increased risk?

Conclusion: It was not unethical for Engineer L to cease work when requested by Client X, without voicing concern about unquantified increased risk. Later, Engineer L did comply with Code provisions that require engineers to notify their employers or clients if a project will not be successful.

Question 2: Would it be ethical for Engineer L to continue working on Client X's project when Client X refuses to invest in the protective measures identified by Engineer L?

Conclusion: It would not be ethical for Engineer L to continue working on Client X's project when Client X refuses to invest in the protective measures identified by Engineer L. Continuing to work on the project after concerns about runoff were quantified would in effect mean Engineer L was placing the clients' financial interest above the engineer's paramount obligation to the public health, safety, and welfare. This, the engineer cannot ethically do.

Code of Ethics References: I.1., I.4., II.1.a., II.3.a., II.3.b., III.1.b., III.3.a.

Case 24-2: Use of Artificial Intelligence in Engineering Practice

Date Published: July 18, 2025

Facts:

Engineer A, an environmental engineer with several years of experience and holding a professional engineering license, was retained by Client W to prepare a comprehensive report addressing the manufacture, use, and characteristics of an organic compound identified as an emerging contaminant of concern. This work required Engineer A to perform an analysis of groundwater monitoring data from a site Engineer A had been observing for over a year. In addition, Engineer A was tasked to develop engineering design documents (plans and specifications) for modifications to groundwater infrastructure at the same site.

Engineer A is known for their strong technical expertise but is, personally, less confident in their technical writing. Previously, Engineer A had relied on guidance and quality assurance reviews by their mentor and supervisor, Engineer B, to refine report drafts. But Engineer B recently retired and was no longer available to Engineer A in a work capacity.

Faced with the need to deliver both the report and the engineering design documents without the review by and mentorship from Engineer B, Engineer A opted to use open-sourced artificial intelligence (AI) software to create an initial draft of the necessary report and to use AI-assisted drafting tools to generate preliminary design documents. The AI drafting software was new to the market and Engineer A had no previous experience with the tool. The AI drafting software was also open-sourced.

For the report, Engineer A gathered the relevant information provided by Client W and relied on the AI software to synthesize the information and generate an initial draft of the necessary report. Engineer A input the information gathered from Client W into the AI software, and, after a few refining prompts, received a first draft of the report generated by the AI software.

Not being familiar with the full functionality of the AI software, including the accuracy and originality of AI-generated text, Engineer A conducted a thorough review of the report, cross-checking key facts against professional journal articles and verifying the phrasing by running search engine queries to ensure the content did not match any existing language. Engineer A also made minor adjustments to some of the wording to personalize the content. Engineer A did not cite their use of AI-software or its large language models, and submitted the draft report to Client W for review, including language to clearly identify that the supplied report was a draft, but applied their seal consistent with state law.

For the engineering design documents, Engineer A entered the information gathered from Client W into the AI software and relied on the AI-assisted drafting tools to

generate a preliminary design of the plans, including basic layouts and technical specifications. Engineer A completed a cursory review of the AI-generated plans and adjusted certain elements to align with site-specific conditions. Again, Engineer A did not cite the AI-assisted drafting tools they used to generate the engineering design documents.

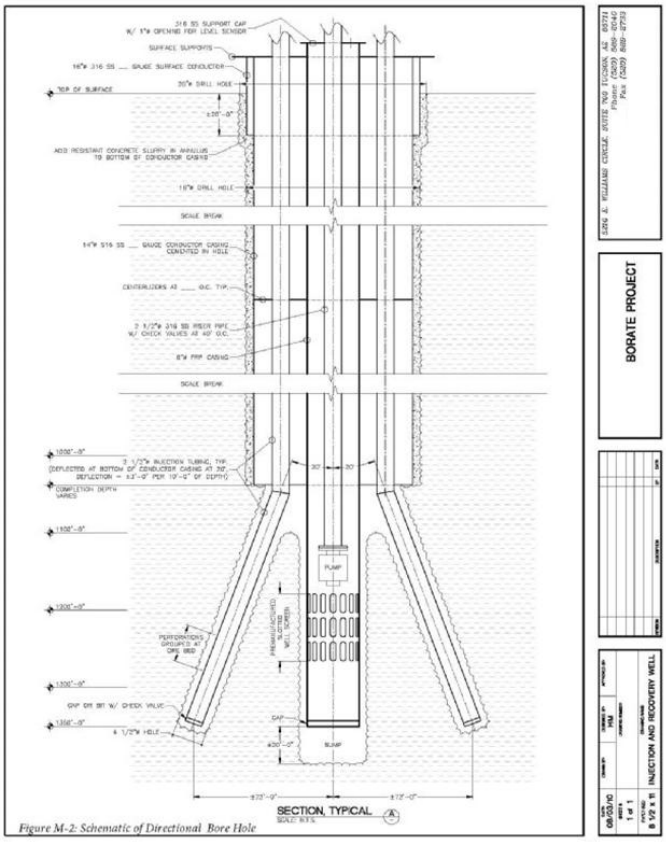


Figure 3: Example groundwater well design plan. Some AI software can assist in creating such plans, referred to as AI-generated plans in this case study.

Source: www.epa.gov/sites/default/files/2020-08/documents/r9uic-ca3-fy19-1-appendix_b-att_l-m_proposed_well_construction_and_diagrams.pdf, p.d.

When Client W reviewed the draft report, Client W noted that the section analyzing the groundwater monitoring data would benefit from minor edits for grammar and clarity, but found the introduction discussing the contaminant's manufacture, use, and characteristics to be exceptionally polished. The Client commented that the report read as if written by two different authors but was otherwise satisfactory. Client W, however, noticed several issues with the AI-generated design documents, including misaligned dimensions and an omission of key safety features required by local regulations. Client W raised concerns about the accuracy and reliability of the engineering design and instructed Engineer A to revise the plans, ensuring that all elements satisfied the necessary professional and regulatory standards.

Question 1: Was Engineer A's use of AI to create the report text ethical, given that Engineer A thoroughly checked the report?

Conclusion: Engineer A's use of AI in report writing was partly ethical, and partly unethical. Engineer A was competent and did thoroughly review and verify the AI-generated content, ensuring accuracy and compliance with professional standards. However, Engineer A did not obtain client permission to disclose private information, nor did Engineer A document required technical citations. Ethical use of AI to create the report text must satisfy all pertinent requirements.

Question 2: Was Engineer A's use of AI-assisted drafting tools to create the engineering design documents ethical, given that Engineer A reviewed the design at a high level?

Conclusion: The use of AI-assisted drafting tools by Engineer A was not unethical per se. However, Engineer A's misuse of the tool, by failing to maintain Responsible Charge over the AI tool and its output before sealing the document and providing it to Client W, was unethical.

Question 3: If the use of AI was acceptable, did Engineer A have an ethical obligation to disclose the use of AI in any form to the Client?

Conclusion: Similar to other software used in the design or detailing process, Engineer A has no professional or ethical obligation to disclose AI use to Client W (unless such disclosure is required under Engineer A's contract with Client W). However, at the time of the BER's review of this case there is no universal guideline mandating AI disclosure in engineering work. Ethical principles favor transparency when AI plays a substantial role in generating work products. To uphold ethical standards, engineers integrating AI into their practice should adopt rigorous verification processes and consider disclosing AI involvement when it plays a significant role in the final product.

Code of Ethics References: I.1., I.2., I.5., II.1.c., II.2.a., II.2.b., III.3., III.8.a., III.9.

Case 24-3: Public Contracting Practices

Date Published: July 18, 2025

Facts:

Engineer A, a licensed professional engineer, joins City D's engineering department as the Assistant City Engineer, responsible for overseeing department staff, managing capital improvement programs, and supervising private development projects. Shortly after starting, Engineer A is approached by Engineer B, also a licensed professional engineer, the owner of a local civil engineering firm, who raises concerns about City D's contracting practices. Engineers A and B have no historical relationship. Engineer B alleges that City D is exclusively awarding traffic engineering contracts to Firm X and civil engineering contracts to Firm Z without utilizing the request for qualifications (RFQ) process. Engineer B expresses frustration that this practice excludes other qualified firms from competing for City D contracts. City D is in a jurisdiction with strong qualification-based selection (QBS) procurement laws at the state and local levels codified within the state's professional engineering licensure laws.



Figure 4: Typical steps in a RFQ and QBS procurement process, as referenced in this case study. Evaluations are based on each firm's qualifications, not on pricing.

Source: Author

Engineer A acknowledges the value of the RFQ and QBS processes in securing engineering services for public entities and commits to investigating Engineer B's concerns. Through the investigation, Engineer A discovers that over the past seven years, City D has awarded consultant contracts exclusively to two firms: Firm X and Firm Z.

City D hired Firm X seven years ago through an RFQ process to maintain the citywide traffic model and review traffic studies submitted by private developers. The contract includes an option for annual extensions for up to 10 years. The work provided by Firm X aligns with the original scope of the RFQ and the resulting contract, with four optional extensions remaining.

City D has entered into five separate contracts with Firm Z over the past six years, with no recorded issues. Records show the first three contracts were awarded through a competitive RFQ process that attracted, on average, four submissions per advertisement. However, the two most recent contracts were awarded solely on the approval of the City Engineer without an RFQ process, even though their dollar amounts exceeded the threshold requiring City D's Council authorization and an RFQ process.

Engineer A brings these findings to City D's Engineer, a licensed professional engineer, recommending improvements to address compliance concerns with the contracting process. City D's Engineer acknowledges that the two most recent contracts with Firm Z did not comply with contracting requirements but dismisses the need for corrective action, citing the convenience and longstanding relationship with Firm Z as justification.

Question 1: Was it ethical for Engineer B to complain to Engineer A?

Conclusion: It was not only ethical for Engineer B to complain to Engineer A, it was ethically required that Engineer B report his belief that statutory obligations were not being followed.

Question 2: Were Engineer A's actions in investigating City D's contracting practices ethical?

Conclusion: It was ethical for Engineer A to investigate City D's contracting practices, both as a part of A's own familiarization process and to follow up on Engineer B's complaints.

Question 3: Because City D's Engineer refuses to change the contract arrangement with Firm Z, what steps must Engineer A take?

Conclusion: Since the City D Engineer indicated they have no plans to change the contract arrangement with Firm Z, Engineer A is obligated to take appropriate action.

Code of Ethics References: I.4., I.6., II.1.e., II.1.f., III.6., III.7., III.8.a.

Case 24-4: Community Engagement for Infrastructure Projects

Date Published: July 18, 2025

Facts:

Engineer M is retained by the City as the lead engineer on a major public infrastructure project to upgrade a highway system that runs through Community P, a historically underserved, underrepresented, and overburdened neighborhood. During the planning phase, the City requires a series of public engagement sessions to gather input from the affected community. Engineer M's firm partners with an engineering consultant, Firm DBA, to coordinate public outreach and conduct these sessions. Firm DBA has a department within its corporate structure that specializes in communications and public relations. It is this department that is directly charged with providing these services on behalf of Firm DBA. Firm DBA has licensed professional engineers in charge or in supervisory roles of all departments in the firm as well as in Firm DBA ownership.

Firm DBA organizes meetings at times and locations that are difficult for Community P residents to attend, such as during work hours at venues far from the area and not easily accessible via public transit. Further, Firm DBA's outreach and notices do not provide for a method for community members to submit written comments. Finally, no virtual meetings are held to gather feedback from interested persons.

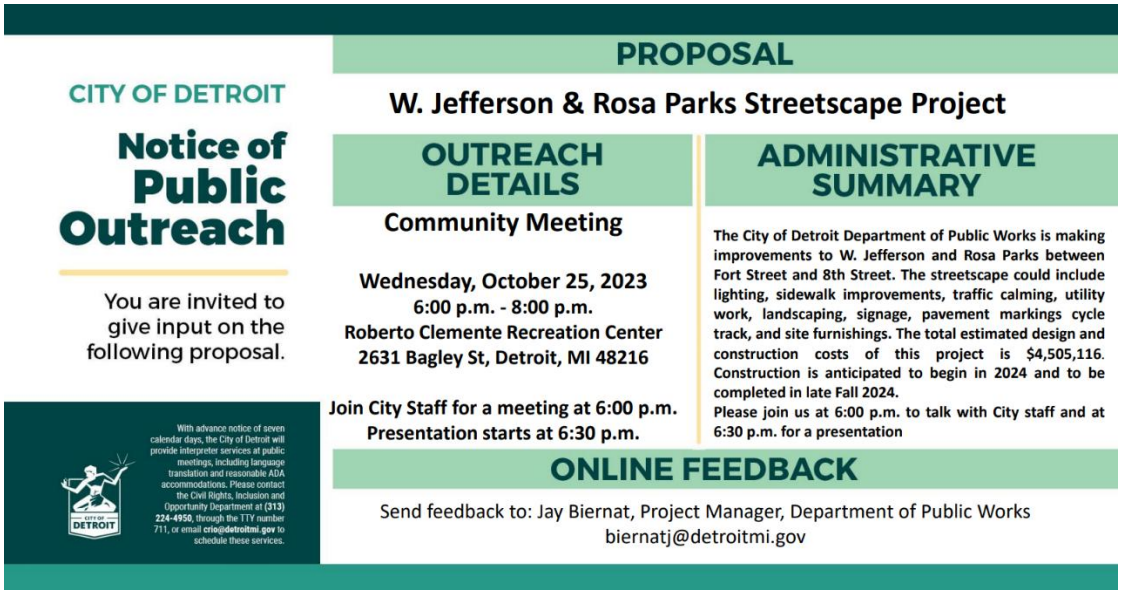


Figure 5: Example Notice of Public Outreach. Unlike the case study, this notice has an email address for written comments, the meeting location is near a bus route, and the meeting room is handicap accessible.

Source: detroitmi.gov/events/notice-public-outreach-w-jefferson-rosa-parks-streetscape-oct-25, p.d.

The area in which Firm DBA holds the public outreach sessions, Community Q, is an alternate route for the major public infrastructure project. Few Community P members participate, and those who do express concerns about the project's potential to displace residents and disrupt local businesses. Many comments are received that support the upgrade through the historically underserved, underrepresented, and overburdened neighborhood (Community P) and not through Community Q.

Engineer M expresses concern to Firm DBA about the public outreach and session locations, but Firm DBA states that these practices are consistent with prior projects undertaken for the City. Further, Firm DBA states that City leaders, citing economic, political, and social considerations, specifically instructed Firm DBA to perform the public engagement sessions in the way that Firm DBA completed them. Despite Engineer M's concerns, Firm DBA prepares a carefully-framed report, omitting details about where the sessions were held and at what time of day the sessions were held, including not identifying that written comments were not accepted. Firm DBA submits a report claiming Community P's support for the project, citing a lack of significant opposition and comments in support of the project.

Question 1: Should Engineer M challenge the validity of Firm DBA's report?

Conclusion: Engineer M should challenge the validity of Firm DBA's report as Firm DBA did not abide by the Code in a number of instances.

Question 2: Should Engineer M raise any concerns with the City, as the client, and, if so, how?

Conclusion: Engineer M should first confer with Firm DBA to correct all discrepancies in the report. If no agreement can be made going forward, Engineer M should confer with the City to outline the ethical obligations. Engineer M should also consider any obligations they may have to report to the state licensure board.

Question 3: Are Firm DBA's actions ethical? Even though Firm DBA is not providing engineering services, are they required to abide by NSPE's Code of Ethics?

Conclusion: The actions of Firm DBA are not ethical under the Code as the services provided were under the supervision and ownership of licensed professional engineers.

Code of Ethics References: I.1., I.3., I.5., II.1.d., II.1.f., II.3.a., III.1.b., III.3.a., III.7.

Case 24-5: Sustainable Development and Resilient Infrastructure

Date Published: July 18, 2025

Facts:

Engineer K, a licensed professional engineer, is hired by the City to design a new flood control system to protect a rapidly growing urban area that has experienced increasingly severe flooding. The City has policies in place to develop new infrastructure projects with resiliency due to climate change in mind. The project's goal is to create a resilient infrastructure that balances immediate protection needs with long-term sustainability.

During the initial design phase, Engineer K identifies two potential approaches, both of which could be successfully designed and implemented:

> **Traditional Approach:** Build a concrete floodwall system to provide immediate protection at a relatively low cost. While effective in the short term, the floodwall system has a high carbon footprint, is prone to deterioration, and may require significant repairs or upgrades within 15 years. Further, the system does not provide for expandability should future flooding risk expand or increase and would require complete demolition and rebuilding if the capacity proved insufficient in the future.
>
> **Sustainable Approach:** Develop a green infrastructure system incorporating wetland restoration and other biodynamic controls. This approach would mitigate flooding while enhancing local biodiversity and reducing carbon emissions. Further, the natural aspects of this approach could readily be expanded if additional capacity is necessary should future flooding risk expand or increase. However, the initial cost is significantly higher than the traditional approach and the system requires several years to fully mature before offering optimal protection.





Figure 5: Top) Concrete flood wall (traditional approach).

Bottom) Wetland and shoreline restoration (sustainable approach).

Sources: commons.wikimedia.org/wiki/File:2024_K%C5%82odzko,_%C5%9Bcianka_przeciwpowodziowa_(2).jpg, Jacek Halicki, CC-BY-SA-4.0; commons.wikimedia.org/wiki/File:120425-A-CE999-006_(6967356336).jpg, p.d.

As part of the project development process, the City directed Engineer K to hold stakeholder meetings to gather feedback on the project. During stakeholder meetings, some commentors expressed a preference for the Traditional Approach due to its lower upfront cost and faster implementation timeline. However, other community and environmental organizations advocated for the Sustainable Approach, citing its long-term environmental and social benefits. Engineer K personally believes the Sustainable Approach aligns better with both City policies and the engineer's professional obligation to promote sustainability and resilience, but recognizes competing priorities of cost, urgency, and long-term impact.

While working on the report and gathering necessary information, Engineer K discovers that the Traditional Approach could disproportionately impact a nearby underserved community by diverting floodwaters to their neighborhood under low-probability but high-volume conditions—particularly if the design capacity of the Traditional Approach is breached. Engineer K presents all available information about both the Traditional Approach and the Sustainable Approach, including the risks and benefits of each approach to the City's leadership during a City Council meeting. The City's leadership decides not to address the identified floodwater issue with the Traditional Approach, ultimately concluding that any action to mitigate the impact on this community would delay the project further and reinforcing the low probability of such conditions occurring. The City approves the Traditional Approach and Engineer K proceeds to work on its implementation.

Question 1: Engineer K personally believes the Sustainable Approach is better. Should Engineer K have only presented information about the Sustainable Approach?

Conclusion: Engineer K should present both approaches to the City if Engineer K believes both are viable solutions.

Question 2: Does Engineer K have any ethical obligations after the City approves the Traditional Approach?

Conclusion: Because Engineer K has entered into a contract to design the new flood water control system, Engineer K has an ethical obligation to act as a faithful agent or trustee. Engineer K is ethically obligated to fulfill their contractual obligations to the City and continue to design the Traditional Approach as approved by the City.

Code of Ethics References: I.1., I.4., II.3.a., II.5.b., III.1.b., III.1.f., III.2.a., III.2.d.

Summary of Ethics References

The following table provides a summary of the code references and ethics topics for each of the five cases covered in this course.

Table 1: Code of Ethics Sections and Topics for the five 2025 BER case studies.

Code of Ethics	Ethics Topic	24-1	24-2	24-3	24-4	24-5
		Stormwater Manag.	AI in Eng. Practice	Public Contracting	Community Engage.	Sustainable Develop.
I.1	Duty to the Public	X	X		X	X
1.2	Competence; Qualifications for Work		X			
1.3	Public Statements and Criticism				X	
I.4	Employer; Faithful Agents and Trustees	X		X		X
1.5	Advertising		X		X	
1.6	Ethical Conduct; Enhance Profession			X		
II.1.a	Duty to the Public	X				
II.1.c	Confidential Information		X			
II.1.d	Associating with Others Firm Name				X	
II.1.e	Unethical Practice by Others			X		
II.1.f	Reporting Ethics Violations			X	X	
II.2.a	Competence; Qualify for Work		X			
II.2.b	Competence; Signing Documents		X			
II.3.a	Misrepresent or Omission of Facts	X			X	X
II.3.b	Opinions	X				
II.5.b	Political Contributions, Gifts, Commissions					X
III.1.b	Duty to Disclose	X			X	X
III.1.f	Harassment and Anti-Discrimination					X
III.2.a	Community Service; Civic Affairs					X
III.2.d	Sustainable Develop.					X

III.3	Avoid Deceiving the Public		X			
III.3.a	Advertising Self-Promotion	X			X	
III.6	Unfair Competition			X		
III.7	Public Statements and Criticism; Unethical Practice by Others			X	X	
III.8.a	Licensure Laws		X	X		
III.9	Credit for Eng. Work; Proprietary Interests		X			

NSPE Statement on Cases

The following is a statement by the NSPE related to the ethics case studies presented in this course.

> The NSPE Board of Ethical Review considers ethical cases involving either real or hypothetical matters submitted to it from NSPE members, other engineers, public officials, and members of the public. The BER reviews each case in the context of the NSPE Code of Ethics and earlier BER opinions. The facts contained in each case do not necessarily represent all of the pertinent facts submitted to or reviewed by the BER and shall not be considered a direct evaluation of any real matter that may have been submitted.
>
> Each opinion is intended as guidance to individual practicing engineers, students, and the public. In regard to the question of application of the NSPE Code of Ethics to engineering organizations (e.g., corporations, partnerships, sole proprietorships, government agencies, and university engineering departments), the specific business form or type should not negate nor detract from the conformance of individuals to the Code. The NSPE Code deals with professional services, which must be performed by real persons. Real persons in turn establish and implement policies within business structures.
>
> This opinion is for educational purposes only and is not legal advice. It may be reprinted without further permission, provided that this statement is included before or after the text of the case and appropriate attribution is provided to the National Society of Professional Engineers' Board of Ethical Review.

NSPE Ethics Case Studies - Quiz

Updated: 11/10/2025

1) Which does NOT normally set behavior standards for professionals?
A. Professional organizations
B. Individuals
C. Employers
D. State regulations

2) The Code of Ethics for Engineers is by which organization?
A. NSPE
B. NCEES
C. NFPA
D. ASME

3) For Case 24-1, it was acceptable to stop work when requested by the client.
A. True
B. False

4) For Case 24-1, why should protective measures be incorporated into the design?
A. Client's request
B. Construction safety
C. Protect public health
D. Reduce liability

5) For Case 24-2, what should be done when using AI-generated content?
A. Review content
B. Cross-check key facts
C. Maintain responsible charge
D. All of the above

6) For Case 24-2, if client gives permission, it is acceptable to upload client data to an AI software.
A. True
B. False

7) For Case 24-3, what practices were considered unethical?
A. Design
B. Procurement
C. Construction
D. Harassment

8) For Case 24-4, whom should the engineer first contact to correct the report?
A. Firm DBA
B. Client
C. Community
D. State board

9) For Case 24-5, the engineer should only present the Sustainable Approach to the client.
A. True
B. False

10) For Case 24-5, after the client choses the Traditional Approach, what should the engineer do?
A. Refuse to do engineering work
B. Insist on the Sustainable Approach
C. Continue with the Traditional Approach
D. Report to State Board